

COMPUTER GRAPHICS (CAC305) LECTURE 17

Er Sanim Kumar Khatri

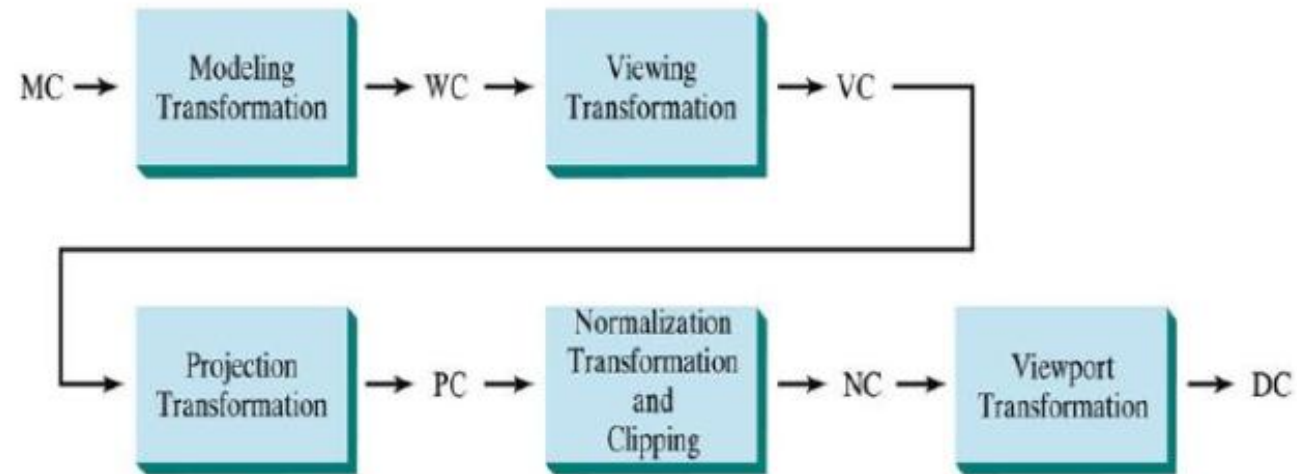
CONTENT

- 4. Visible Surface Determination and computer graphics algorithm
 - 4.1 Image space and object space techniques
 - 4.2 Hidden surface removal – Depth comparison
 - 4.3 Z-buffer Algorithm
 - 4.4 Back Face Detection
 - 4.5 The painter's algorithm
 - 4.6 Scan-Line Algorithm
 - 4.7 Light and Color different color model (RGB, CMY, YIQ)

Three Dimensional Viewing

- In two-dimensional graphics applications, viewing operations transfer positions from the world-coordinate plane to pixel positions in the plane of the output device. Using the rectangular boundaries for the world-coordinate window and the device viewport, a two-dimensional package maps the world scene to device coordinates and clips the scene against the four boundaries of the viewport.
- For three-dimensional graphics applications, the situation is a bit more involved, since we now have more choices as to how views are to be generated.
- First of all, we can view an object from any spatial position: from the front, from above, or from the back.
- Or we could generate a view of what we would see if we were standing in the middle of a group of objects or inside a single object, such as a building.
- Additionally, three-dimensional descriptions of objects must be projected onto the flat viewing surface of the output device.

3D Viewing Pipeline



- To take a snapshot, we first need to position the camera at a particular point in space, then need to decide on the camera orientation. (i.e., which way do we point the camera and how should we rotate it around the line of sight to set the up direction for the picture). Finally, when we snap the shutter, the scene is cropped to the size of the shutter, the scene is cropped to the size of the 'window' (aperture) of the camera and light from the visible surface is projected.
- Each model or object has its own dimension and coordinate system. It is called modeling coordinate. Modeling transformation is to take all the objects in a single scene by using transformation such translation, rotation etc. To set object is called modeling translation.

Viewing Pipeline

- After modeling translation, the objects come to a scene or a coordinate system is called world coordinate.
- To set the camera on some position, angle or orientation is called viewing transformation. From viewing transformation we get viewing coordinate.
- Projection transformation is to adjust focus, zoom in, zoom out etc. Projection transformation creates projection coordinates.
- Workstation or viewport translation is just like to click the button to save the image in the device.
- Once the scene has been modeled, world coordinate positions are converted to viewing coordinate.
- The VC system is used in graphics system as a reference for specifying the observer viewing position and the position of the projection plane analogous to camera film plane.
- Projection operations are performed to convert VC description of a scene to co-ordinate positions on the projection plane.
- The projection plane is then mapped to output device.

Viewing Pipeline



Viewing co-ordinates

Suppose that the viewing co-ordinates are specified in world co-ordinates. We need to transform each vertex specified in world co-ordinates to view co-ordinates.

1. Translate viewing co-ordinate origin to world co-ordinates origin.
2. Apply rotation to align u , v and n with the world x , y and axes.

Transformation matrices:

Translation

If the view reference point is specified at world position (X_0, Y_0, Z_0) this point is translated to world origin with the matrix transformation.

$$\begin{bmatrix} 1 & 0 & 0 & -x_0 \\ 0 & 1 & 0 & -y_0 \\ 0 & 0 & 1 & -z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation

The composite rotation matrix for viewing transformation is then

$$\begin{bmatrix} u_1 & u_2 & u_3 & 0 \\ v_1 & v_2 & v_3 & 0 \\ n_1 & n_2 & n_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Projection

Representing an n-dimensional object into an n-1 dimension is known as projection.

It is process of converting a 3D object into 2D object, we represent a 3D object on a 2D plane $\{(x,y,z) \rightarrow (x,y)\}$.

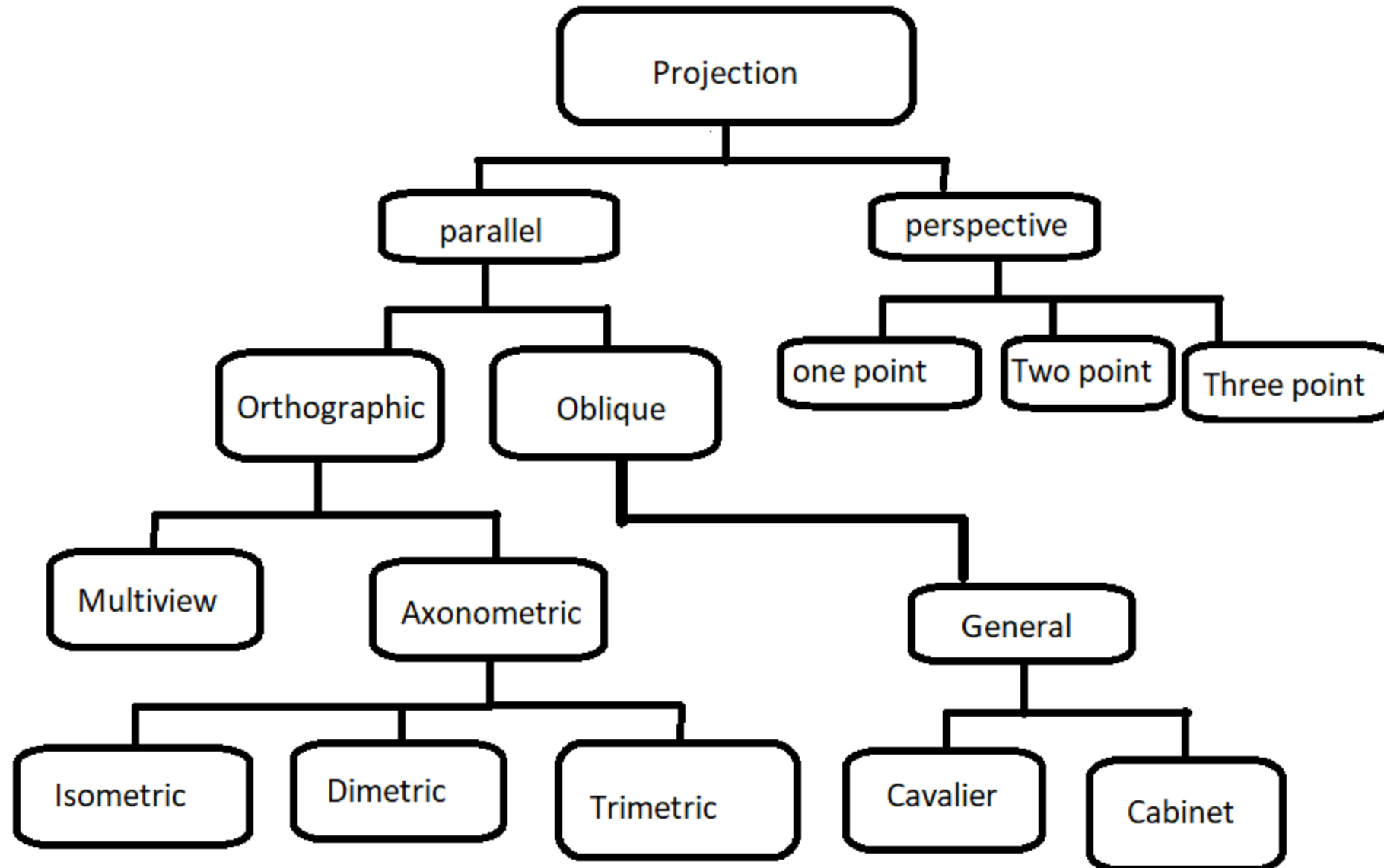
It is also defined as mapping or transforming of the object in projection plane or view plane.

When geometric objects are formed by the intersection of lines with a plane, the plane is called the projection plane and the lines are called projections.

Types of Projections:

- Parallel projections
- Perspective projections

Projection



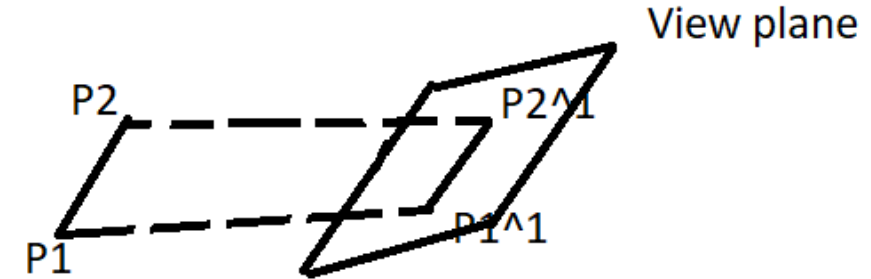
Parallel Projection

Co-ordinate positions are transformed to the view plane along parallel lines.

A projection is said to be parallel, if center of projectic is at an infinite distance from the projected plane.

A parallel projection preserves relative proportion of objects, accurate views of the various sides of an object are obtained with a parallel projection.

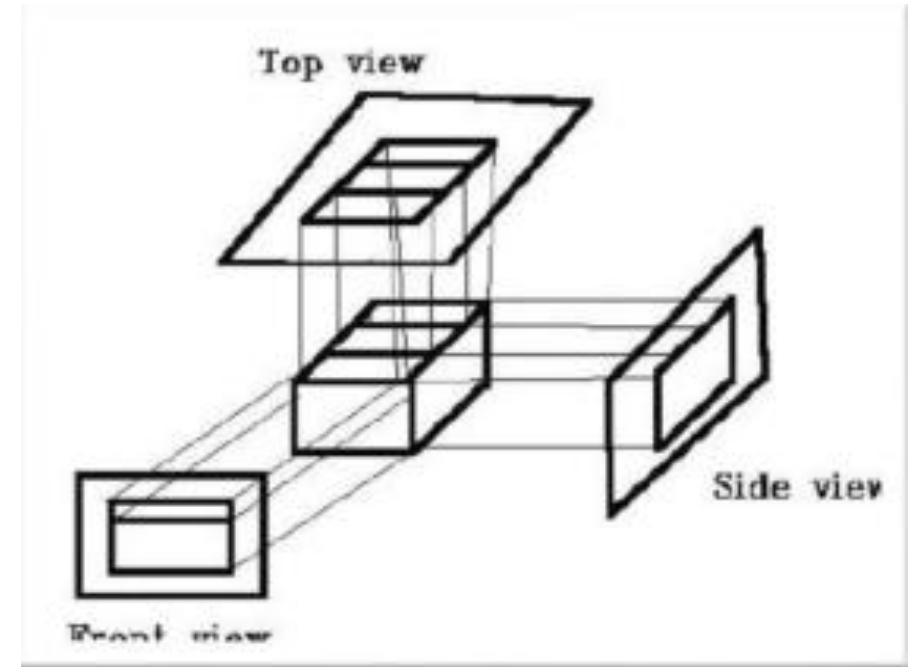
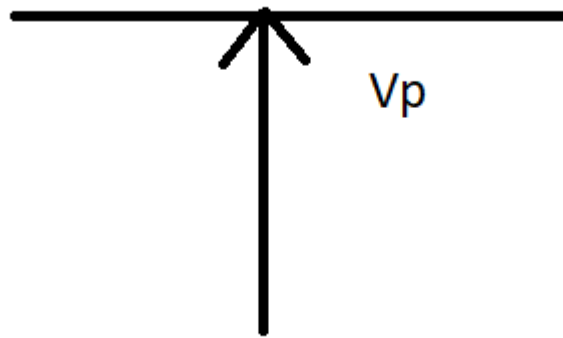
It preserves relative propositions of objects, and it is used in drafting to produce scale drawings of 3D objects.



Orthographic Projection

In orthographic projection, the center of projection lies at infinity and the projections are perpendicular to the view plane. So, a true size and shape of a single face of object is obtained. In orthographic projection the direction of projection is normal to the projection of the plane. There are three types of orthographic projections –

- Front Projection
- Top Projection
- Side Projection

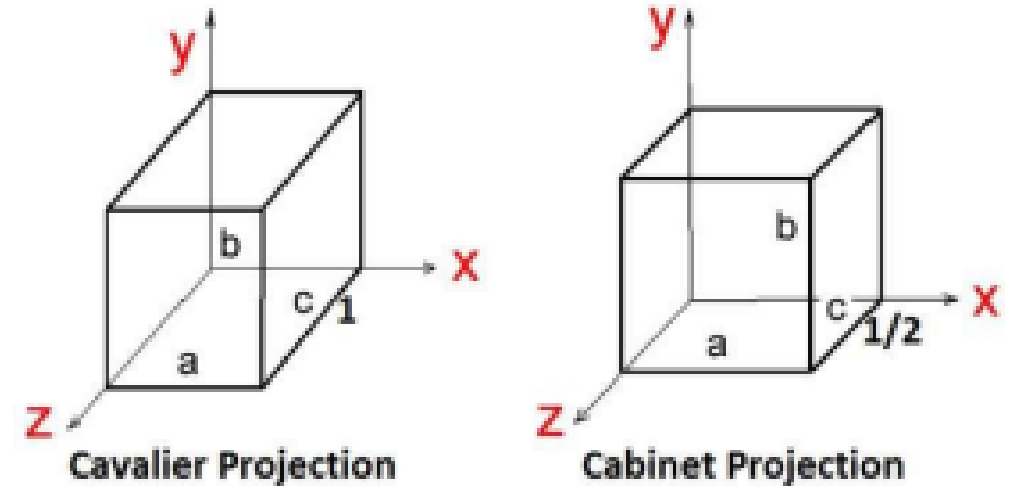


Oblique Projection

A projection in which the angle between the projectors and the plane of projection is not equal to 90 degree is called oblique projection.

An oblique projection is formed by parallel projections from a center of projection at infinity that intersects the plane of projection at an oblique angle.

There are two types of oblique projections - **Cavalier** and **Cabinet**.

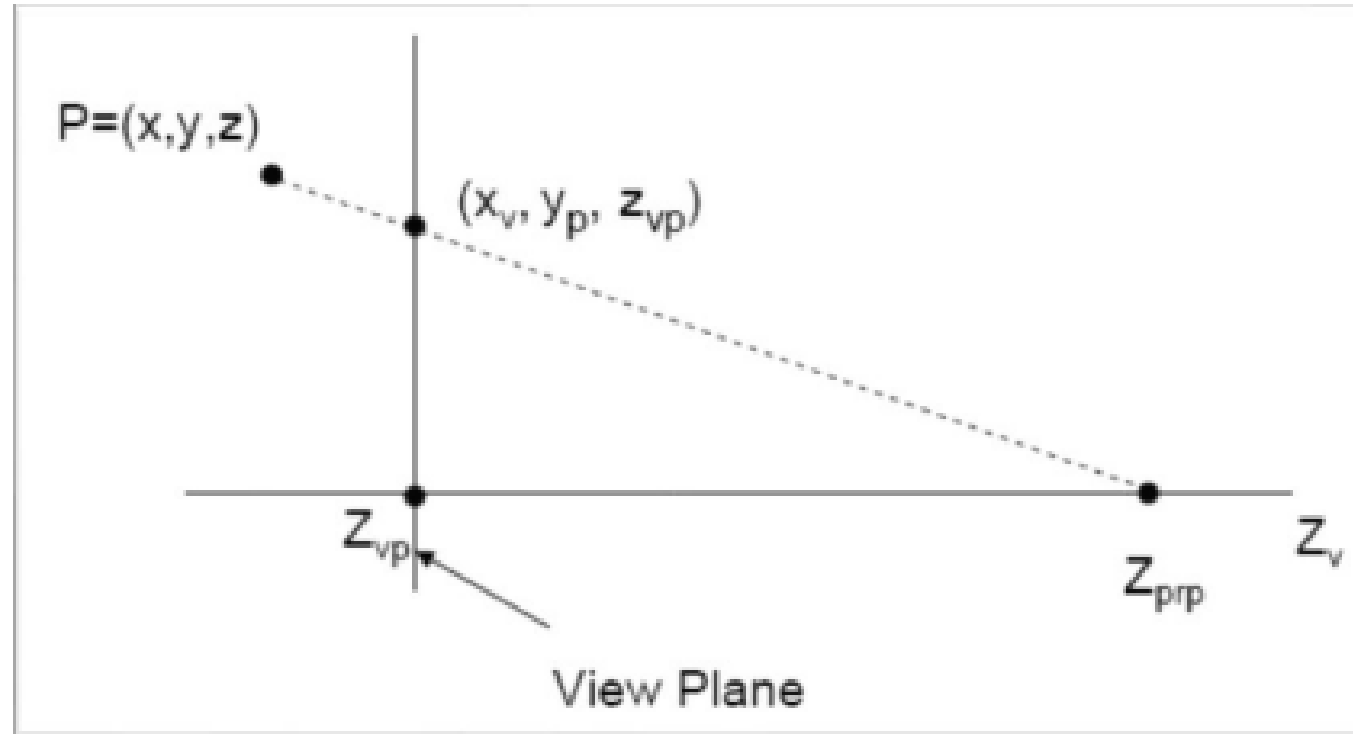


Perspective projections

- A perspective projection is the one produced by straight lines radiating from a common point and passing through point on the sphere to the plane of projection.
- Perspective projection is a geometric technique used to produce a three dimensional graphic image on a plane, corresponding to what person sees.
- Any set of parallel lines of object that are not parallel to the projection plane are projected into converging lines. A different set of parallel lines will have a separate vanishing point.
- Coordinate positions are transferred to the view plane along lines that converge to a point called projection reference point.
- The distance and angles are not preserved and parallel lines do not remain parallel. Instead, they all converge at a single point called center of projection there are 3 types of perspective projections.

Two characteristic of perspective are vanishing point and perspective force shortening. Due to fore shortening objects and lengths appear smaller from the center of projections. The projections are not parallel and we specify a center of projection cop.

Perspective projections



Thank you

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Hidden surface removal

Visible surface detection or Hidden surface removal is major concern for realistic graphics for identifying those parts of a scene that are visible from a chosen viewing position. Several algorithms have been developed. Some require more memory, some require more processing time and some apply only to special types of objects.

Visible surface detection methods are broadly classified according to whether they deal with objects or with their projected images.

These two approaches are :

- **Object-Space methods:** Compares objects and parts of objects to each other within the scene definition to determine which surface as a whole we should label as visible.
- **Image-Space methods:** Visibility is decided point by point at each pixel position on the projection plane.

Hidden surface removal

The Hidden surface approaches discussed here are:

- **Back-Face Detection**
- **Depth Buffer Method**
- **A-buffer method**
- **Scan Line Method**
- **Depth Sorting Method**

Hidden surface removal

DEPTH-BUFFER METHOD

Depth Buffer Method is the commonly used image-space method for detecting visible surface. It is also known as z-buffer method. It compares surface depths at each pixel position on the projection plane. It is called z-buffer method since object depth is usually measured from the view plane along the z-axis of a viewing system.

Each surface of scene is processed separately, one point at a time across the surface. The method is usually applied to scenes containing only polygon surfaces, because depth values can be computed very quickly and method is easy to implement. This method can be applied to non-planar surfaces.

DEPTH-BUFFER METHOD

Algorithm: Z-buffer

1. Initialize depth buffer and refresh buffer so that for all buffer position (x,y)
 $\text{depth}(x,y) = \infty$, $\text{refresh}(x,y) = I_{\text{background}}$.
2. For each position on each polygon surface, compare depth values to previously stored value in depth buffer to determine visibility.
 - Calculate the depth Z for each (x,y) position on polygon
 - If $Z < \text{depth}(x,y)$
then $\text{depth}(x,y) = Z$
 $\text{refresh}(x,y) = I_{\text{surface}}(x,y)$

Hidden surface removal

DEPTH-BUFFER METHOD

Where,

$I_{\text{background}}$ is the intensity value for background

$I_{\text{surface}}(x,y)$ is intensity value for surface at pixel position (x,y) on projected plane.

After all surface are processed, the depth buffer contains the depth value of the visible surface and refresh buffer contains the corresponding intensity values for those surface. The depth value of the surface position (x,y) are calculated by plane equation of surface

$$Z = (-Ax - By - D)/C$$

Let Depth Z at position $(x+1, y)$

$$Z' = (-A(x+1) - By - D)/C$$

$$Z' = Z - (A/C)$$

– (A/C) is constant for each surface so succeeding depth value across a scan line are obtained from preceding values by simple calculation.

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A-BUFFER METHOD

The A buffer method represents an antialiased, area-averaged, accumulation-buffer method developed by Lucasfilm for implementation in the surface-rendering system called REYES (an acronym for "Renders Everything You Ever Saw")

The A-buffer method expands the depth buffer so that each position in the buffer can reference a linked list of surfaces. Thus, more than one surface intensity can be taken into consideration at each pixel position, and object edges can be antialiased.

Hidden surface removal

A-BUFFER METHOD

Each position in the A-buffer has two fields:

- Depth field: stores a positive or negative real number
- Intensity field: stores surface-intensity information or a pointer value

If the depth field is positive, the number stored at that position is the depth of a single surface overlapping the corresponding pixel area. The intensity field then stores the RCB components of the surface color at that point and the percent of pixel coverage.

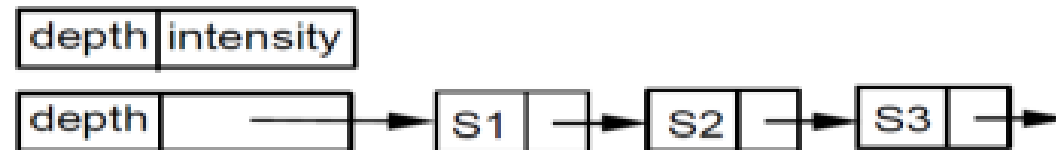
If the depth field is negative, this indicates multiple-surface contributions to the pixel intensity. The intensity field then stores a pointer to a linked list of surface data

Hidden surface removal

A-BUFFER METHOD

Data for each surface in the linked list includes:

- RGB intensity components
- opacity parameter (percent of transparency)
- Depth
- percent of area coverage
- surface identifier
- other surface-rendering parameters
- pointer to next surface



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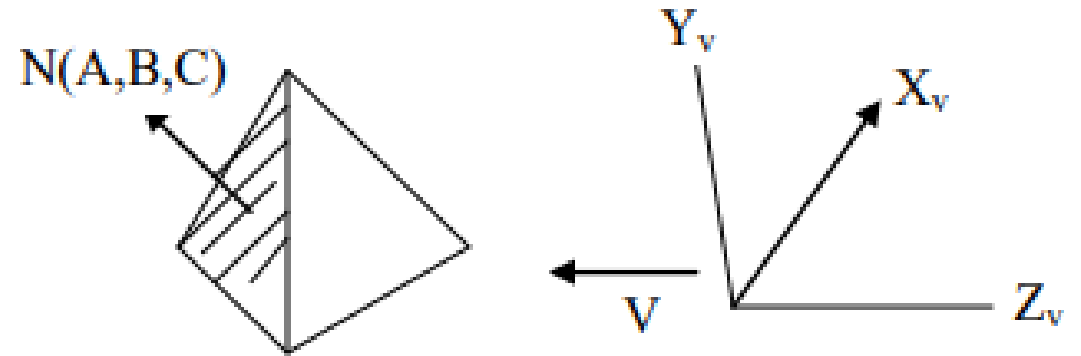
Hidden surface removal

BACK-FACE DETECTION

A fast and simple object space method used to remove hidden surface from a 3D object drawing is known as "Plane equation method" and applied to each side after any rotation of the object takes place. It is commonly known as back-face detection of a polyhedron is based on the "inside-outside" tests.

A point (x, y, z) is inside a polygon surface if
 $Ax + By + Cz + D = 0$

We can simplify this test by considering the normal vector N to a polygon surface which has Cartesian components (A, B, C) . If V is the vector in viewing direction from the eye position then this polygon is a back face if, $V \cdot N > 0$



Hidden surface removal

In the equation:

$$Ax + By + Cz + D = 0$$

if A,B,C remains constant, then varying value of D results in a whole family of parallel planes.

One of which ($D = 0$) contains the origin of the co-ordinates system and ,

If $D > 0$, plane is behind the origin(Away from observer)

If $D < 0$, plane is in front of origin(towards the observer)

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DEPTH SORTING METHOD

This method uses both object space and image space method. In this method the surface representation of 3D object are sorted in of decreasing depth from viewer.

The conceptual steps that performed in depth-sort algorithm are

1. Sort all polygon surface according to the largest (farthest) Z co-ordinate of each.
2. Resolve any ambiguity this may cause when the polygons Z extents overlap, splitting polygons if necessary.
3. Scan convert each polygon in ascending order of larger Z-co-ordinate i.e. farthest surface first (back to front)

In this method, the newly displayed surface is partly or completely obscure the previously displayed surface. Essentially, we are sorting the surface into priority order such that surface with lower priority (higher z, far objects) can be obscured by those with higher priority (low z-value).

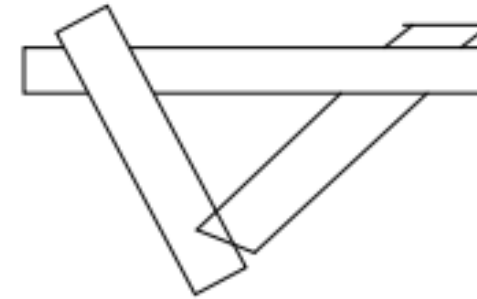
Hidden surface removal

DEPTH SORTING METHOD

This algorithm is also called "Painter's Algorithm" as it simulates how a painter typically produces his painting by starting with the background and then progressively adding new (nearer) objects to the canvas.

Problem: One of the major problem in this algorithm is intersecting polygon surfaces. As shown in fig. below.

- Different polygons may have same depth.
- The nearest polygon could also be farthest.



We cannot use simple depth-sorting to remove the hidden-surfaces in the images. Solution: For intersecting polygons, we can split one polygon into two or more polygons which can then be painted from back to front. This needs more time to compute intersection between polygons. So it becomes complex algorithm for such surface existence.

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Hidden surface removal

SCAN LINE METHOD

This is image-space method for removing hidden surface which is extension of the scan line polygon filling for polygon interiors. Instead of filling one surface we deal with multiple surface here.

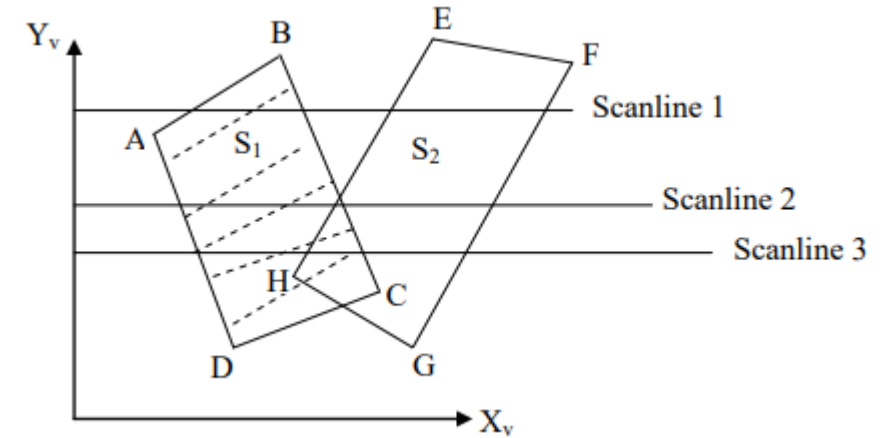
As each scan line is processed, all polygon surface intersecting that line are examined to determine which are visible. We assume that polygon table contains the co-efficient of the plane equation for each surface as well as vertex edge, surface information, intensity information for the surface, and possibly pointers to the edge table.

Hidden surface removal

SCAN LINE METHOD

In figure above, the active edge list for scan line 1 contains information from edge table for edge AB, BC, EH, FG.

- For positions along this scan line between edge AB and BC, only the flag for surface S1 is on
- Therefore no depth calculation must be made using the plane coefficients for two surfaces and intensity information for surface S1 is entered from the polygon table into the refresh buffer.
- Similarly between EH&FG. Only the flag for S2 is on. No other positions along scan line 1 intersect surface. So intensity values in the other areas are set to background intensity
- For Scanline 2 & 3, the active edge list contains edges AD, EH BC, FG. Along scanline 2 from edge AD, to edge EH only surface flag for S1 is on, but between edges EH& BC, the flags for both surface is on. In this interval, depth calculation is made using the plane coefficients for the two surfaces.



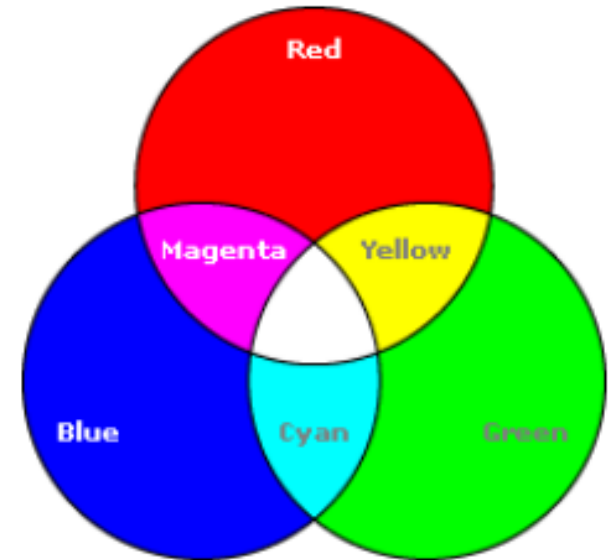
Color Models

A color model is simply a way to define color. A model describes how color will appear on the computer screen or on paper. Two popular color models are:

- **RGB** (Red, Green, Blue)
- **CMYK** (Cyan, Magenta, Yellow, Black)
- **YIQ**

1. RGB (Red, Green, Blue)

The RGB model is used when working with screen-based designs. A value between 0 and 255 is assigned to each of the light colors, Red, Green, and Blue. So for example, if we wanted to create a purely blue color, Red would have a value of 0, Green would have a value of 0 and Blue would have a value of 255 (pure blue). To create black, Red, Green and Blue would each have a value of 0 and to create white, each would have a value of 255. RGB is known as an “additive” model and is the opposite of the subtractive color model.

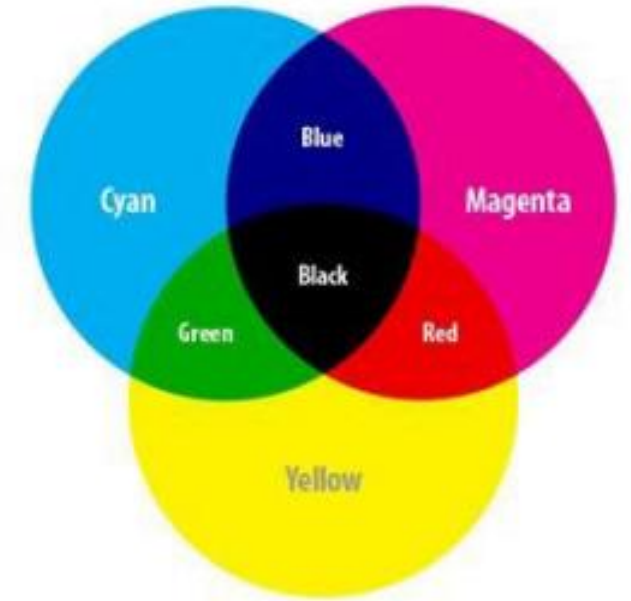


Color Models

2. CMYK (Cyan, Magenta, Yellow, Black).

The CMYK model is used for print work and it describes colors based on their percentage of Cyan, Magenta, Yellow and Black. These four colors are used by commercial printers and bureaus and you may also find that your home printer uses these colors too. These four colors are needed to reproduce full color artwork in magazines, books and brochures. By combining Cyan, Magenta, Yellow and Black on paper in varying percentages, the illusion of lots of colors is created.

CMYK is known as a “subtractive” color model. White is the natural color of the paper or other background, while black results from a full combination of colored inks.



Color Models

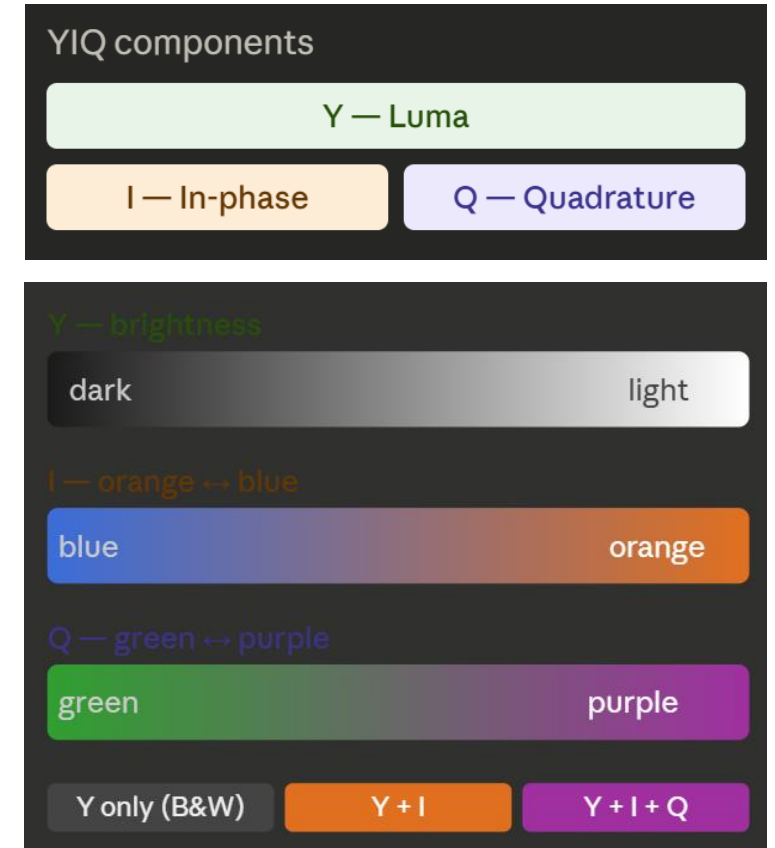
3. YIQ (Luma, In-phase, Quadrature)

The YIQ model is used in analog television broadcasting, particularly in the NTSC system used in North America and Japan. It encodes color information by separating brightness from color data.

YIQ splits a color signal into three components: Y (luma: the brightness or luminance), I (in-phase: the orange–blue color axis), and Q (quadrature: the green–purple color axis).

This separation was designed so that black-and-white televisions could use only the Y channel, while color TVs decoded all three. The human eye is more sensitive to detail in the I channel than in Q, so I was given a wider bandwidth during transmission.

YIQ is considered a "luma-chroma" model. Unlike RGB, which treats all channels equally, YIQ prioritizes perceptual brightness, making it well suited for image compression and video encoding where human visual sensitivity matters most.



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Illumination Theory and Models

Illumination Theory and Models

Realistic displays of a scene are obtained by generating perspective projections of objects and by applying natural lighting effect to the visible surfaces.

Illumination model is used to calculate the intensity of light that we should see at a given point on the surface of an object. An illumination model (equation) expresses the components of light reflected from or transmitted (refracted) through a surface. There are three basic light components: ambient, diffuse and specular.

Light sources:

Total reflected light from an opaque non luminous object, is the sum of the contributions from light sources and other reflecting light surfaces in the scene. So, a surface that is not directly exposed to a light source may still be visible if nearby objects are illuminated. Light emitting sources are light bulbs, sun etc. Light reflecting sources are walls of a room, other reflecting surfaces etc. A luminous object, in general, can be both a light source and a light reflector, e.g. a plastic globe with a light bulb, etc.

Light sources

Point source:

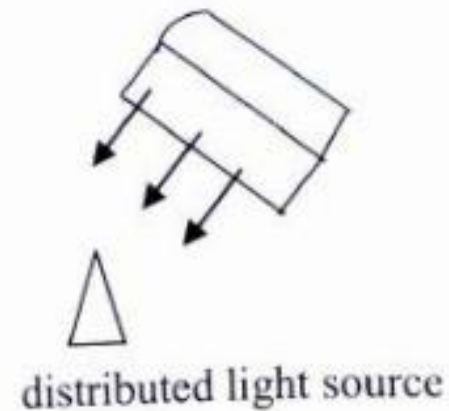
Simplest model for a light emitter

Rays from the source then follow radially diverging paths from the source position. e.g. light bulb.

Distributed light source:

Area of the source is not small compared to the surface in the scene.

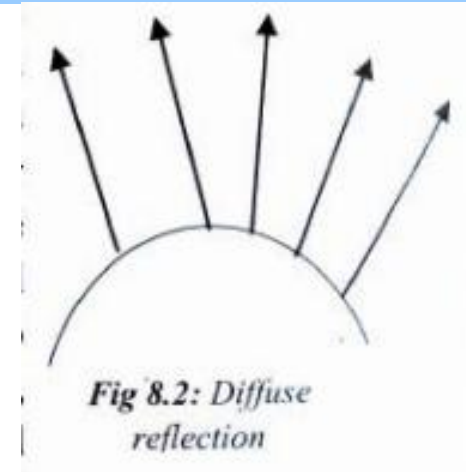
A nearby source, such as the long fluorescent light is more accurately modeled as a distributed light source.



Light reflection

Diffuse reflection:

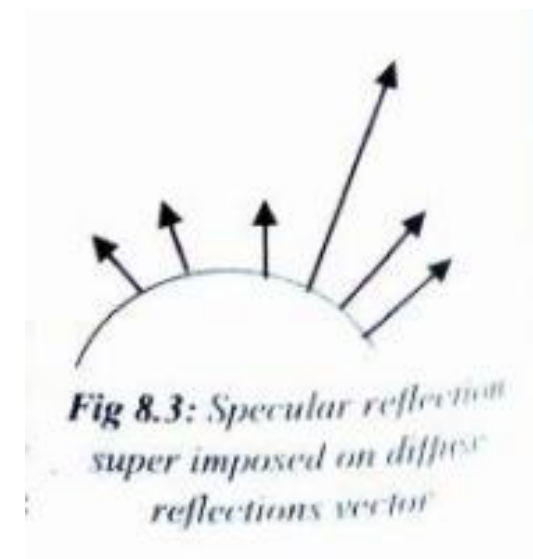
Diffuse reflection is the reflection of light from a surface such that a ray incident on the surface is scattered at many angles rather than at just one angle as in the case of specular reflection. An ideal diffuse reflecting surface is said to exhibit Lambertian reflection, meaning that there is equal luminance when viewed from all directions lying in the half-space adjacent to the surface.



Specular reflection:

Specular or regular reflection, is the mirror-like reflection of waves, such as light, from a surface. In this process, each incident ray is reflected at the same angle to the surface normal as the incident ray, but on the opposing side of the surface normal in the plane formed by incident and reflected rays. The result is that an image reflected by the surface is reproduced in mirror-like (specular) fashion.

The law of reflection states that for each incident ray the angle of incidence equals the angle of reflection, and the incident, normal, and reflected directions are coplanar.



Basic Illumination Models

Illumination model is method for calculating light intensities. Light calculations are based on the optical properties of the surface, the back ground lighting conditions and the lights ource specifications. Optical parameters are used to set surface properties such as transparency, opacity etc, and these control the amount of reflection and absorption of incident light. All light sources are considered to be point source specified with a coordinate position an intensity value (color).

Ambient light reflection:

This is a simplest illumination model. We can think of this model, which has no external light source-self-luminous objects. A surface that is not exposed directly to light source still will be visible if nearby objects are illuminated.

The combination of light reflections form various surfaces to produce a uniform illumination is called ambient light or background light.

Ambient light has no spatial or directional characteristics and amount on each object is a constant for all surfaces and all directions.

Basic Illumination Models

So if I_a is the amount of ambient light incident on any surface, the ambient light reflection is given by ambient illumination equation,

$$I = I_a K_a$$

We have K_a = ambient reflectivity or ambient reflection coefficient which ranges from 0 to 1. It is a material property.

Diffuse reflection:

Ambient light reflection is an approximation of global diffuse lighting effects.

Diffuse reflections are constant over each surface in a scene, independent of viewing direction.

Amount of incident light that is diffusively reflected is defined with a surface parameter K_d called diffuse reflection coefficient or diffuse-reflectivity.

K_d is assigned a constant value in the interval 0 to 1 according to the reflecting properties we want the surface to have.

For highly reflecting surface, $K_d \rightarrow 1$ and for a very dull surface $K_d \rightarrow 0$

Diffuse reflection intensity at any point on the surface if exposed only to ambient light is:

$$I_{\text{ambdiff}} = K_d * I_a$$

Basic Illumination Models

Assuming diffuse reflections from the surface are scattered with equal intensity in all directions, independent of the viewing direction (surface called. "Ideal diffuse reflectors") also called Lambertian reflectors and governed by Lambert's cosine law:

$$I_{\text{diff}} = K_d I_l \cos\theta$$

Where I_l is the intensity of the point light source.

A surface is illuminated by a point source only if the angle of incidence is in

the range 0° to 90° for which $\cos$ is in the range 0 to 1. When $\cos$ is negative, the light source is behind the surface. If N is the unit normal vector to a surface and L is the direction vector to the point source from a position on the surface then,

$$N \cdot L = |N| |L| \cos\theta$$

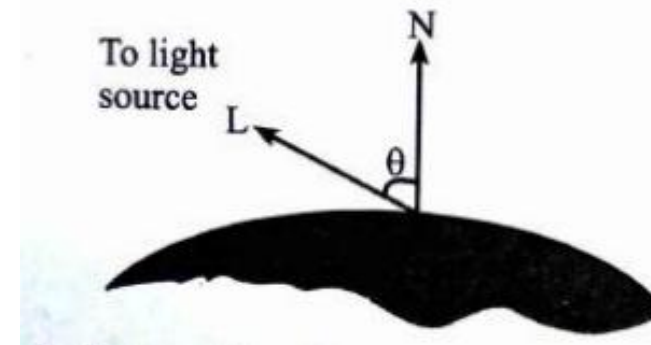
For unit vector $|N|$ and $|L| = 1$

So, $N \cdot L = \cos\theta$

$$I_{\text{diff}} = K_d I_l (N \cdot L)$$

In addition, many graphics packages introduce an ambient reflection coefficient K_a to modify the ambient-light intensity I_a

$$I_{\text{diff}} = I_a K_a + K_d I_l \cos\theta$$



Basic Illumination Models

Specular Reflection and the Phong model:

When we look at an illuminated shiny surface, such as polished metal, a person's forehead, we see a highlight or bright spot, at certain viewing direction. Such phenomenon is called specular reflection. It is the result of total or near total reflection of the incident light in a concentrated region around the "specular reflection angle = angle of incidence".

N - unit vector normal to surface at incidence point

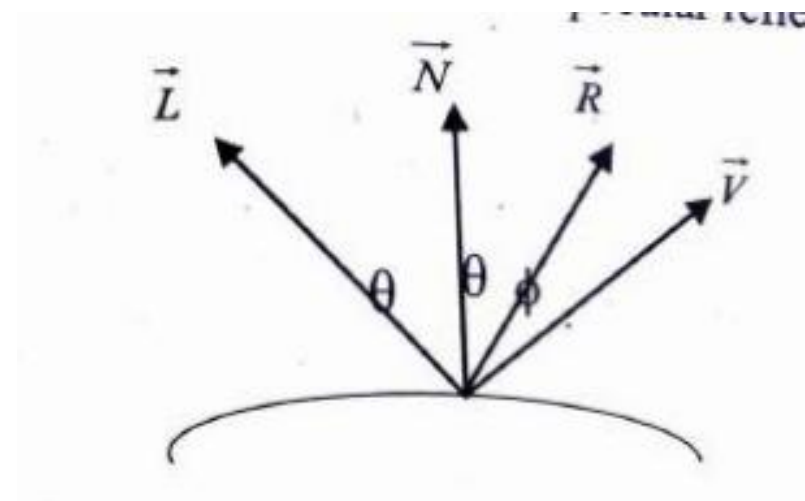
R - unit vector in the direction of ideal specular reflection.

L - unit vector directed to words point light source.

V - unit vector pointing to the viewer from surface.

Φ - the viewing angle relative to the specular reflection direction.

For ideal reflector (perfect mirror), incident light is reflected only in the specular reflection direction. V & R coincide ($\Phi = 0$)



Shiny surfaces have a narrow specula-reflection range, whereas dull surfaces have a wider reflection range.

An empirical model for calculating specular-reflection range developed by Phong Bui Tuong-called "Phong specular reflection model (or simply Phong model), sets the intensity of specular reflection proportional to $\cos^{n_s} \Phi$ where Φ ranges from 0 to 90 .

Specular reflection parameter n_s is determined by type of surface.

Basic Illumination Models

Very shiny surface with large value n_s (say 100 or more) and dull surface, smaller n_s (down to 1). For a perfect reflector, n_s is infinite.

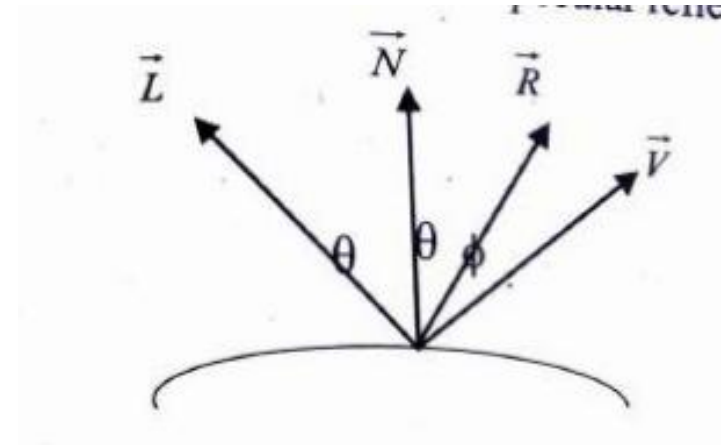
Phong model calculate the specular reflection light intensity as

$$I_{\text{spec}} = W(\theta) I_i \cos^{n_s} \Phi$$

Where I_i = incident light intensity

$W(\theta) = 1$ at $\theta = 90^\circ$, and all of the incident light is reflected.

$W(\theta)$ = Specular-reflection coefficient



Many opaque objects exhibit almost constant specular reflection for all incidence angles. For this case, we can replace $W(90)$, with a constant specular-reflection coefficient k_s , whose value can be assigned in the range 0 to 1.

$V \cdot R = |V| |R| \cos \Phi$, since it is unit vector

$$I_{\text{spec}} = K_s I_i (V \cdot R)^{n_s}$$

Combined diffuse and specular reflection,

$$\begin{aligned} I &= I_{\text{diff}} + I_{\text{spec}} C \\ &= I_a K_a + K_d I_i \cos \theta + K_s I_i (V \cdot R)^{n_s} \end{aligned}$$

Polygon Rendering Methods

Illumination model is applied to fill the interior of polygons.

Two ways of polygon surface rendering

1. Single intensity for all points in a polygon
2. Interpolation of intensities for each point in a polygon

Methods:

1. Constant Shading
2. Gouraud Shading
3. Phong Shading
4. Fast Phong Shading

Constant Shading

It is also called flat shading.

It is fast and simple.

A single intensity is calculated for each polygon and all points over the surface of the polygon are then displayed with the same intensity value.

Useful for quickly displaying the general appearance of a curved surfaces.

Procedure:

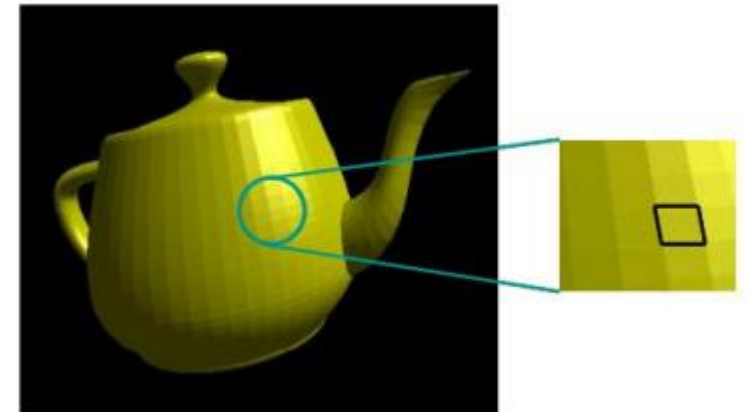
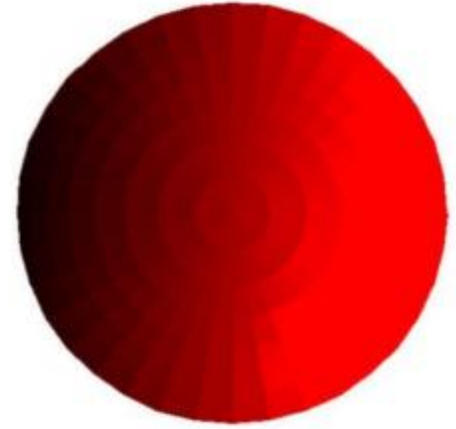
1. Take a point on the object surface and calculate the intensity.
2. Render the surface with the same intensity throughout the surface.
3. Repeat above steps for each polygon surface.

Assumptions:

1. Object is polyhedron.
2. All light sources should be sufficiently far from the surface (i.e N.L and attenuation function are constant over the polygon surfaces)
3. Viewing position is sufficiently far (i.e V.R is constant over the surface)

Drawback:

The sharp intensity discontinuation is seen in the border between two polygons.



Gouraud Shading

It uses the intensity interpolation method.

Renders a polygon surface by linearly interpolating intensity values across the surface.

Intensity discontinuity at the edges of polygons is eliminated by matching intensity values of each polygon with adjacent polygons thus eliminating the intensity discontinuation that can occur in flat shading.

Produces more realistic results, but requires consider more calculations.

Calculation Steps:

- Determine the average unit normal vector at each polygon vertex
- Calculate each of the vertex intensities by applying an illumination model
- Linearly interpolate the vertex intensities over the polygon surface

Drawback:

Mach bands: bright and dark intensity streaks caused by linear interpolation of intensities

- Could be reduced by dividing the surface into large number of polygons or by using other methods, such as Phong shading

Phong Shading (Normal Vector Interpolation shading)

More accurate method for rendering

Fundamental: Interpolate normal vectors and apply illumination model to each surface point

Calculation steps:

1. Determine average unit normal vectors at each polygon vertex

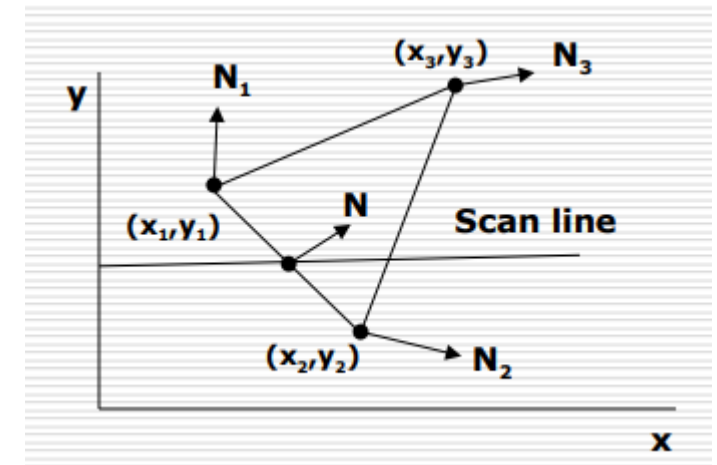
$$N_v = \frac{\sum_{k=1}^n N_k}{\left| \sum_{k=1}^n N_k \right|}$$

2. Linearly interpolate the vertex normals over the surface of the polygon

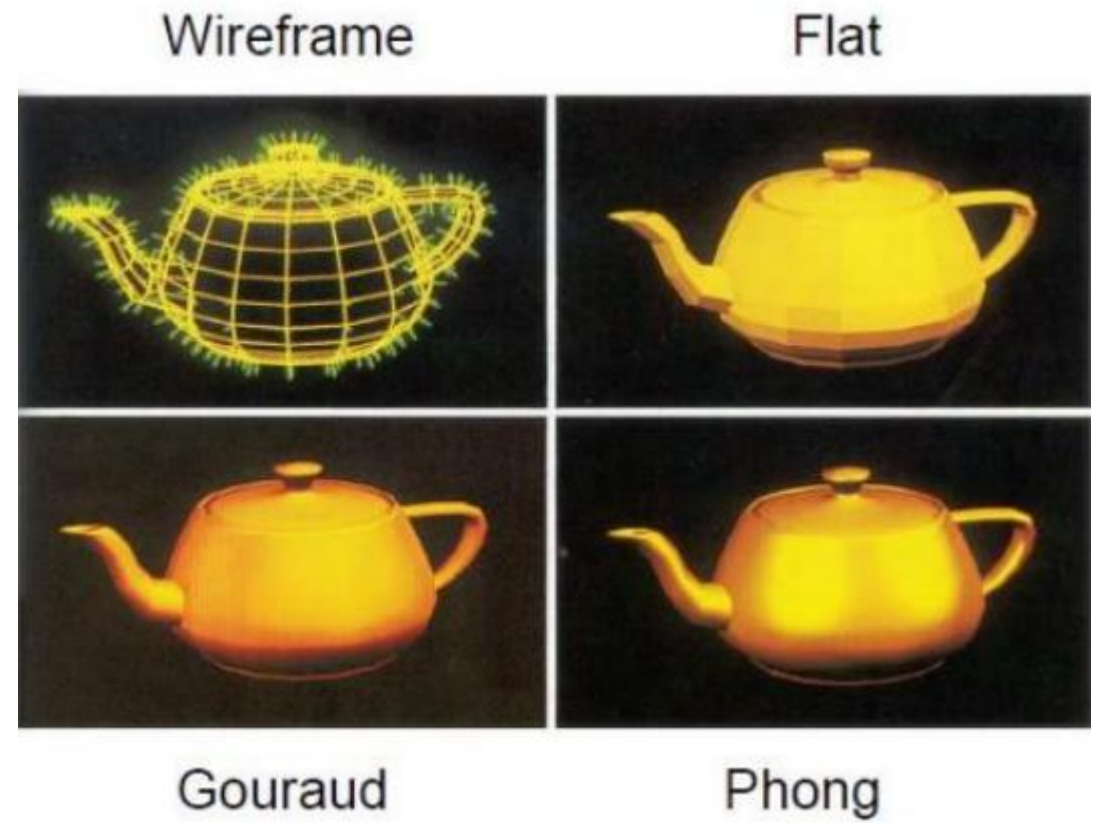
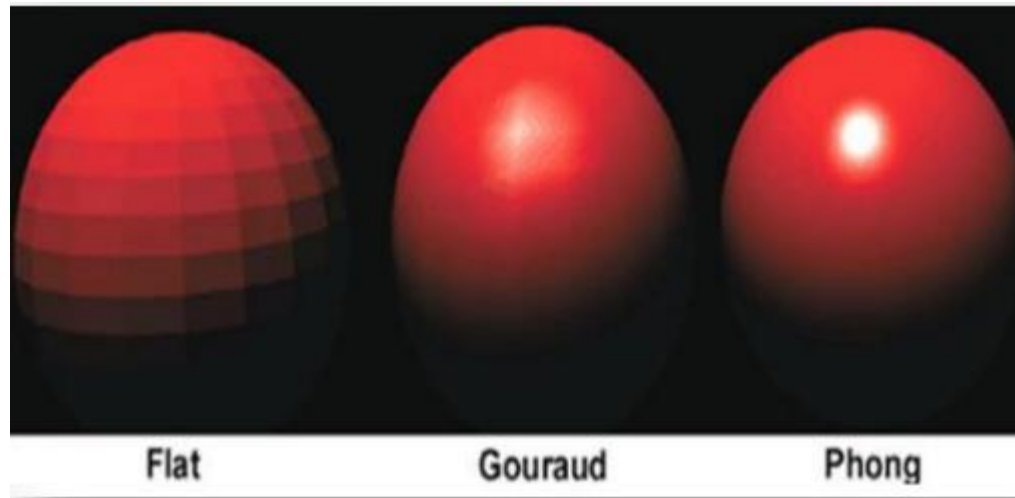
$$N = \frac{y - y_2}{y_1 - y_2} N_1 + \frac{y_1 - y}{y_1 - y_2} N_2$$

3. Apply an illumination model along each scan line to calculate projected pixel intensities for the surface points

Trade-off: requires considerably more calculations



Shading



Thank you